

Outline for Final, Autumn-24

	Structure of matter
1.	Definition: Crystalline and non-crystalline solid, single crystal and polycrystalline solids, lattice, basis, lattice point, Crystal, unit cell.
2.	Crystal systems and their classification
3.	Coordination number (NaCl and CsCl)
4.	Packing fraction (definition + SC, BCC, FCC calculation)
5.	Crystal planes (ex: 111...etc and directions, Miller indices,
6.	Relation between inter-planar spacing & Miller indices. (Expression)
7.	All expression related math
	Defects & Bond in Solid
1.	Crystal defects (Definition+ classification+ some concept of different types of crystal defects)
2.	Crystal bonding (Definition+ classification+ some concept of different types of crystal bonding)
x 3.	calculation of cohesive and bonding energy
4.	Distinction between metal, insulator and semiconductor in terms of energy band.
5.	All expression related math
	Relativity
1.	Inertial frame , Non Inertial frame
2.	Postulates of special theory of relativity
3.	Lorentz transformation (Only Equation)
x 4.	Variation of Mass with Velocity expression
5.	Time dilation (Full expression)
6.	Length contraction
7.	Mass-energy Relation
8.	All expression related math
	Modern Physics
1.	Bohr's atomic model Postulates
2.	Radius and total energy of Hydrogen atom
3.	Atomic nucleus and properties (Nuclear Mass, Nuclear Size, Nuclear density, Nuclear charge)
4.	Definition of : Mass defect, Binding energy, Binding energy per nucleon Also, Calculate the following properties of A_ZX particle...ex: ${}^6_6C^{12}$: or any i. Nuclear Mass ii. Nuclear Size iii. Nuclear Density iv. Nuclear Charge v. Nuclear Mass defect vi. Nuclear Binding Energy vii. Nuclear Binding Energy per nucleon.
x 5.	Photo-electric effect + Einstein Photo-electric effect equation
6.	Compton Effect ..only concept
x 7.	De-Broglie waves.... Only concept
8.	X-ray. Properties of X-ray, uses of X-ray, Bragg;s law and its expression
9.	Atomic spectra and Zeeman effect..... Only concept

10.	All expression related math
	Radioactivity
1.	Definition, Radioactive elements , properties of radioactive radiation
2.	Disintegration law + expression of Disintegration law
3.	Half life + expression of half life
4.	Mean life + expression of mean life
5.	alpha decay, beta decay, gamma decay
6.	nuclear fission & fusion
7.	Nuclear cross section
8.	All expression related math